

# Leave Balance Converter

## Technical Overview & Migration Handover

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### 1. The Problem We Are Solving

Clients currently managing leave balances within Simplifi without an API connection face a painful three-step manual process:

- Export a leave balance report from their payroll system (Smartly or IMS/MYOB).
- Manually reformat and remap the data — renaming columns, converting units, removing redundant rows — in Excel or a text editor.
- Re-upload the transformed file into Simplifi.

This is error-prone, time-consuming, and entirely avoidable. Each manual transformation step introduces the risk of incorrect mappings, wrong unit conversions (hours vs days), or stale data being imported.

*On the question of API connectivity: Smartly and IMS/MYOB do not currently expose public APIs to Simplifi for leave balance data. While a direct integration would be the ideal long-term solution, it is simply not available today. In the meantime, expecting clients to absorb a more manual or degraded workflow is not a realistic option — particularly when competing platforms already offer seamless leave balance imports. This tool bridges that gap now, with no burden placed on the client.*

### 2. What the Tool Does

The Leave Balance Converter is a lightweight, single-page web application that eliminates the manual transformation step entirely. A payroll administrator:

- Selects their payroll provider (Smartly or IMS/MYOB).
- Drops or browses to their exported CSV leave balance file.
- Chooses optional output settings (skip zero balances, select specific leave codes).
- Reviews a live preview table and downloads a Simplifi-ready CSV in one click.

The entire workflow takes under 60 seconds and requires no technical knowledge.

### 3. Technical Architecture

The application is intentionally simple by design — a deliberate architectural choice to maximise security, minimise maintenance overhead, and eliminate infrastructure dependencies.

|                        |                                                                                                                             |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Type</b>            | Static single-page application (SPA)                                                                                        |
| <b>Language</b>        | Vanilla HTML5, CSS3, and JavaScript — no framework, no build step                                                           |
| <b>File count</b>      | One file: index.html (all logic, styles, and markup are self-contained)                                                     |
| <b>Dependencies</b>    | None — zero external libraries or npm packages                                                                              |
| <b>Data processing</b> | All CSV parsing and data transformation runs in the user's browser via the Web APIs (FileReader, Blob, URL.createObjectURL) |
| <b>Server</b>          | None — the application has no backend, no API calls, and no server-side logic                                               |
| <b>Browser support</b> | All modern browsers (Chrome, Edge, Firefox, Safari)                                                                         |

Provider support is implemented as a plug-in pattern inside the JavaScript. Each provider has:

- A `detect()` function that reads the first few lines of the uploaded CSV to identify the format automatically.
- A `parse()` function that reads the provider-specific CSV structure into a normalised row array.
- A `convert()` function that maps the normalised rows into Simplifi's import column format.

Current providers: Smartly, IMS/MYOB. Crystal Payroll is stubbed and marked "Coming Soon" in the UI.

### 4. Security & Data Privacy

Data privacy was a primary design constraint. The following guarantees are built into the architecture:

*No employee data — names, IDs, balances, or any other personal information — ever leaves the user's device. The application does not transmit, store, log, or cache any data from uploaded files.*

|                         |                                                                                                                                                                              |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Network requests</b> | Zero. The page makes no HTTP calls after initial load. There is no analytics, no telemetry, no third-party scripts.                                                          |
| <b>PII ingestion</b>    | None. File contents are read by the browser's FileReader API and processed entirely in JavaScript memory. Data is never written to disk, a database, or any remote endpoint. |
| <b>Data retention</b>   | None. All processed data lives in JavaScript variables and is discarded when the tab is closed or the user clicks "Start over".                                              |
| <b>Authentication</b>   | Not required. The tool is intentionally public-facing within the Simplifi ecosystem — access control is handled at the hosting layer.                                        |
| <b>File upload</b>      | No upload occurs. The browser reads the file locally via the File API. The "upload" is a local file-read operation only.                                                     |
| <b>Dependencies</b>     | No third-party JavaScript libraries are loaded, eliminating supply-chain risk entirely.                                                                                      |

*The footer of the application displays the following notice, visible to all users:*

*"All data is processed locally in your browser — nothing is sent to any server."*

## 5. Current Deployment

The tool is currently hosted via GitHub Pages from a personal GitHub repository. Details:

|                      |                                                                                                                                                 |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Repository</b>    | <a href="https://github.com/Jamie-Callawick/Leave-Balance-Worker-App">https://github.com/Jamie-Callawick/Leave-Balance-Worker-App</a>           |
| <b>Owner</b>         | Personal account — Jamie Callawick (Jamie-Callawick)                                                                                            |
| <b>Branches</b>      | main (production), Dev (development / staging)                                                                                                  |
| <b>Hosting</b>       | GitHub Pages — served as a static site directly from the repository                                                                             |
| <b>URL format</b>    | <a href="https://jamie-callawick.github.io/Leave-Balance-Worker-App/">https://jamie-callawick.github.io/Leave-Balance-Worker-App/</a> (current) |
| <b>Build process</b> | None — changes pushed to main are immediately live. No CI/CD pipeline is currently configured.                                                  |
| <b>Access</b>        | Publicly accessible via the GitHub Pages URL                                                                                                    |

## 6. Migration: Personal GitHub to Simplifi Organisation

Moving the repository from a personal account to the Simplifi GitHub organisation is a straightforward process. The steps below assume the Simplifi org already exists on GitHub. No code changes are required.

|   |                                 |                                                                                                                                                                                                                                                                     |
|---|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Create org repository</b>    | Create a new repository called Leave-Balance-Worker-App (or equivalent) inside the Simplifi GitHub organisation. Set visibility to match your policy (private recommended during transition).                                                                       |
| 2 | <b>Transfer or mirror</b>       | Option A (recommended): Settings > Danger Zone > Transfer ownership > select the Simplifi org. Preserves all git history, branches, and issues. Option B: Clone and push to a new org repo — use only if a clean slate is preferred.                                |
| 3 | <b>Update GitHub Pages</b>      | In the new org repository, go to Settings > Pages > set Source to the main branch (root). The public URL will become <a href="https://simplifi.github.io/Leave-Balance-Worker-App/">https://simplifi.github.io/Leave-Balance-Worker-App/</a> (adjust for org name). |
| 4 | <b>Update any shared links</b>  | If the old GitHub Pages URL has been shared with clients or in documentation, update references to the new URL. GitHub Pages does not redirect the old URL after a transfer.                                                                                        |
| 5 | <b>Branch protection</b>        | Apply branch protection rules to main: require pull request reviews before merge, disable force-push. This prevents accidental direct pushes to production.                                                                                                         |
| 6 | <b>Team access</b>              | Add relevant Simplifi dev team members with appropriate roles (Write or Maintain). Remove Jamie's personal admin access if the account will no longer manage the repo.                                                                                              |
| 7 | <b>Custom domain (optional)</b> | To use a branded URL (e.g. <a href="https://tools.simplifi.work/leave-converter">tools.simplifi.work/leave-converter</a> ), add a CNAME file to the repo and configure the DNS record to point to github.io per GitHub Pages documentation.                         |
| 8 | <b>Verify &amp; smoke test</b>  | After migration, load the new URL, upload a sample Smartly and IMS/MYOB CSV, and confirm the download output is correct. Check browser console for any errors.                                                                                                      |

## 7. Risks & Mitigations

The following risks have been identified across hosting, security, and ongoing maintenance. Each is rated by likelihood and impact.

| Risk                                         | Likelihood | Impact | Mitigation                                                                                                                                                                                                                                              |
|----------------------------------------------|------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GitHub Pages outage</b>                   | Low        | Medium | GitHub Pages has a strong uptime record. If availability is a concern, the single index.html file can be mirrored to any static host (Netlify, Cloudflare Pages, Azure Static Web Apps) with minimal effort.                                            |
| <b>Personal account access lost</b>          | Low        | High   | The repository sits under a personal GitHub account. If access is lost before migration, the codebase must be re-created from a local copy. Mitigate by completing the migration to the Simplifi org promptly and keeping a local backup of index.html. |
| <b>No uptime SLA</b>                         | Medium     | Low    | GitHub Pages does not offer an SLA. For an internal-use tool this is acceptable. If an SLA becomes a requirement, move hosting to a managed platform such as Azure Static Web Apps.                                                                     |
| <b>Tool URL changes post-migration</b>       | High       | Medium | After migrating to the Simplifi org the GitHub Pages URL will change, breaking any existing bookmarks or documentation links. Communicate the new URL to all users before cutover and update all internal documentation.                                |
| <b>Scope creep into PII handling</b>         | Low        | High   | The current tool handles only leave balance figures. If future requirements add employee names or IDs to the output, a privacy review is required before release to ensure the no-PII guarantee remains valid.                                          |
| <b>No dedicated maintainer post-handover</b> | Medium     | Medium | If the dev team does not assign an owner, bug reports or provider format changes (e.g., Smartly updates their CSV export) will go unaddressed. Assign a named maintainer and add the repo to standard team monitoring.                                  |
| <b>Publicly accessible URL</b>               | Medium     | Low    | The tool is accessible to anyone with the link. No sensitive data is processed, so this is low risk. If policy requires restricted access, Cloudflare Access or a GitHub org-only Pages restriction can be applied without code changes.                |
| <b>No automated tests</b>                    | Medium     | Medium | There are no unit or integration tests. A provider format change could silently break conversions. Add a GitHub Actions workflow that runs sample CSV conversions and validates the output on every pull request.                                       |

## 8. Future Considerations

The following enhancements are suggested as the tool matures and API availability changes:

- Other payroll support — there is a placeholder provider slot already stubbed in the code; only the detect/parse/convert functions need to be implemented once a sample export is available.
- Direct API integration — if Smartly or IMS/MYOB release API access to leave balance data in the future, the transformation logic in this tool can be reused server-side to enable a true one-click sync.
- CI/CD pipeline — a simple GitHub Actions workflow can lint, validate, and optionally run smoke tests on pull requests before merge.
- Version pinning — embedding a version number in the UI footer (e.g., v1.2.0) will help support staff identify which version a client is running when troubleshooting.
- Access control — if the tool needs to be restricted to Simplifi clients only, authentication can be added at the CDN/hosting layer (e.g., Cloudflare Access) without changing any application code.

## 9. Point of Contact

For questions about the codebase, the current deployment, or the migration process, contact:

|                       |                            |
|-----------------------|----------------------------|
| <b>Owner / Author</b> | Jamie Callawick            |
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| <b>GitHub</b>         | github.com/Jamie-Callawick |

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